

JEE MAIN REVISION MASTER: COMPOUNDS CONTAINING NITROGEN

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 229 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Basicity of Amines & Gabriel Phthalimide Synthesis	175 PYQs	Amine Basics
Very High	Diazonium Salts & Carbylamine Test	55 PYQs	Diazonium & Tests

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Basicity of Amines & Gabriel Phthalimide Synthesis	$\text{Basicity in water: } (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N} > \text{NH}_3$ $(\text{CH}_3)_3\text{N} > (\text{NH}_3) \text{ in gas phase}$
Diazonium Salts & Carbylamine Test	$\text{Diazotization: } \text{Ar-NH}_2 + \text{NaNO}_2 + 2\text{HCl} \rightarrow \text{Ar-N}_2^+ \text{Cl}^- + \text{NaCl} + 2\text{H}_2\text{O}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Basicity of Amines & Gabriel Phthalimide Synthesis (Very High Yield — 175 PYQs)

Core Principle: Interplay of inductive, steric, and solvation factors on amine basicity in gas vs aqueous phases, and Hoffmann bromamide degradation.

■ **Caution / JEE Pitfall:** In gas phase, basicity order is purely inductive: $3^\circ > 2^\circ > 1^\circ > \text{NH}_3$. In aqueous solution, solvation and steric factors disrupt this order.

■ **Speed Trick:** Gabriel phthalimide synthesis yields PURE aliphatic primary amines (1°); aromatic amines cannot be prepared because aryl halides do not undergo $\text{S}_\text{N}2$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Which statements are True? A. In Hoffmann bromamide degradation, 4 moles of NaOH and 2 moles of Br_2 are consumed per mole of an amide B..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Arrange the following compounds according to increasing order of boiling points. $n\text{-C}_4\text{H}_9\text{OH}$ (A), $n\text{-C}_4\text{H}_9\text{NH}_2$ (B), $n\text{-C}_4\text{H}_{10}$ (C)..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Product C of the following reaction sequence will be..." [Ans: N/A]

3.2 Diazonium Salts & Carbylamine Test (Very High Yield — 55 PYQs)

Core Principle: Preparation of benzene diazonium chloride ($0-5^{\circ}\text{C}$), Sandmeyer/Gattermann halogenation, azo coupling dyes, and Hinsberg separation.

■ **Caution / JEE Pitfall:** Carbylamine test gives foul-smelling isocyanide ONLY with primary amines (1°); secondary and tertiary amines give NO reaction.

■ **Speed Trick:** Hinsberg reagent ($\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$): 1° amine gives product soluble in KOH; 2° amine gives product insoluble in KOH; 3° amine does not react.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Given below are two statements : In the light of the above statements, choose the correct answer from the options given below..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#2 (MCQ):** "Which of the following amino acid will give violet coloured complex with neutral ferric chloride solution?..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#2 (MCQ):** "A student has been given a compound 'x' of molecular formula- $\text{C}_{(6)}\text{H}_{(7)}\text{N}$. 'x' is sparingly soluble in water. However, on additio..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

Arrange the following compounds according to increasing order of boiling points.

$\text{n-C}_{(4)}\text{H}_{(9)}\text{OH}$ (A), $\text{n-C}_{(4)}\text{H}_{(9)}\text{NH}_{(2)}$ (B), $\text{n-C}_{(4)}\text{H}_{(10)}$ (C) and $\text{C}_{(2)}\text{H}_{(5)}\text{NHC}_{(2)}\text{H}_{(5)}$ (D).

- (A) $\text{C} < \text{B} < \text{A} < \text{D}$
- (B) $\text{D} < \text{C} < \text{B} < \text{A}$
- (C) $\text{C} < \text{D} < \text{B} < \text{A}$
- (D) $\text{D} < \text{B} < \text{A} < \text{C}$

Correct Answer: None

Solution Analysis: (A) $\text{n-C}_{(4)}\text{H}_{(9)}\text{OH}$ (an alcohol) forms strong intermolecular hydrogen bonding between O-H groups of different molecules. Because of this strong attraction, more heat is needed to separate the molecules, so its boiling point is the highest. (B) $\text{n-C}_{(4)}\text{H}_{(9)}\text{NH}_{(2)}$ (a primary amine) also shows intermolecular hydrogen bonding due to the presence of N-H bonds. But nitrogen is less electronegative than oxygen, so N-H hydrogen bonding is weaker than O-H hydrogen bonding. Hence, its boiling point is lower than the alcohol. (C) $\text{n-C}_{(4)}\text{H}_{(10)}$ is a non-polar hydrocarbon. It cannot form hydro

Benchmark Problem 2: JEE Main 2026 (Online) 24th January Evening Shift (Q#3)

A student has planned to prepare acetanilide from aniline using acetic anhydride. The student has started from 9.3 g of aniline. However, the student has managed to obtain 11 g of dry acetanilide. The % yield of this reaction is :-

- (A) 59.5%
- (B) 72.5
- (C) 97.5
- (D) 81.5%

Correct Answer: None

Solution Analysis: Reaction (1:1): $\text{C}_6\text{H}_5\text{NH}_2 + (\text{CH}_3\text{CO})_2\text{O} \rightarrow \text{C}_6\text{H}_5\text{NHCOCH}_3 + \text{CH}_3\text{COOH}$ Molar mass of aniline $\text{C}_6\text{H}_7\text{N} \approx 93 \text{ g mol}^{-1}$ $n(\text{aniline}) = (9.3 / 93) = 0.10 \text{ mol}$ Molar mass of acetanilide $\text{C}_8\text{H}_9\text{NO} \approx 135 \text{ g mol}^{-1}$ Theoretical mass: $m_{\text{(theor)}} = 0.10 \times 135 = 13.5 \text{ g}$ Percent yield: $\% \text{yield} = (11 / 13.5) \times 100 \approx 81.5\%$ Answer: Option D (81.5%)